Unit -3: Performance Metrics and Network Optimization

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1. Measuring Network Performance

a. Latency

- **Definition:** Time taken for a data packet to travel from source to destination.
- **Types:**
  - *Propagation Delay:* Time for signal to travel through the medium.
  - *Transmission Delay:* Time to push bits onto the wire.
  - *Processing Delay:* Time to process packet headers.
  - *Queuing Delay:* Time a packet waits in queue before transmission.

b. Bandwidth

- **Definition:** Maximum rate at which data can be transferred over a network path.
- **Units:** Bits per second (bps), Mbps, Gbps.
- **Theoretical vs. Effective Bandwidth:**
  - *Theoretical:* Maximum possible.
  - *Effective:* Real-world bandwidth after overheads.

c. Throughput

- **Definition:** Actual amount of data successfully transmitted over a network per unit time.
- **Affected by:** Protocol overheads, congestion, retransmissions.

Practical Example:

If a 10 MB file takes 10 seconds to transfer:

Throughput=10 MB10 sec=1 MBps=8 Mbps\text{Throughput} = \frac{10 \text{ MB}}{10 \text{ sec}} = 1 \text{ MBps} = 8 \text{ Mbps}Throughput=10 sec10 MB=1 MBps=8 Mbps

Tools to Measure Performance:

- *Ping, Traceroute, iPerf, Wireshark.*

2. Load Balancing Techniques and Traffic Engineering

a. Load Balancing

- **Purpose:** Distribute network traffic evenly to avoid congestion and improve performance.
- **Techniques:**

- *Round Robin:* Sequentially cycles through paths.
- *Least Connections:* Chooses the path/server with the fewest active connections.
- *Hash-based:* Based on source/destination IP or port.

b. Load Balancers

- **Types:**
  - *Hardware-based:* Dedicated appliances (e.g., F5). https://www.f5.com/products/big-ip-services/iseries-appliance
  - *Software-based:* Open-source or virtual (e.g., HAProxy, Nginx). https://www.openlogic.com/blog/nginx-vs-haproxy

c. Traffic Engineering

- **Definition:** Optimization of operational performance of a network by dynamically analyzing, predicting, and regulating data flow.
- **Used in:** MPLS networks, SD-WAN.
- **Goals:**
  - Minimize delay.
  - Optimize bandwidth usage.
  - Avoid congestion points.

Example:

MPLS Traffic Engineering routes data along pre-defined paths avoiding bottlenecks, enhancing performance over traditional IP routing.

3. Fault Tolerance and Redundancy in High-Performance Networks

a. Fault Tolerance

- **Definition:** Ability of a network to continue operating despite failures.
- **Achieved by:**
  - *Redundant hardware*
  - *Failover systems*
  - *Dynamic routing protocols* (e.g., OSPF, EIGRP)

b. Redundancy

- **Types:**
  - *Link Redundancy:* Multiple paths between nodes (e.g., dual links).

- *Device Redundancy:* Backup routers/switches (e.g., HSRP, VRRP).
- *Power Redundancy:* UPS, redundant PSUs.

c. Protocols Supporting High Availability

- *HSRP (Hot Standby Router Protocol)*
- *VRRP (Virtual Router Redundancy Protocol)*
- *GLBP (Gateway Load Balancing Protocol)*

Example Scenario:

Two routers configured with HSRP; if primary fails, secondary takes over IP without affecting users.

Tutorial Task:

1. Use ping and traceroute to compare latency across local and global websites.
2. Setup a basic load balancing configuration using HAProxy on Linux.
3. Simulate link failure in a Packet Tracer topology and observe OSPF failover.